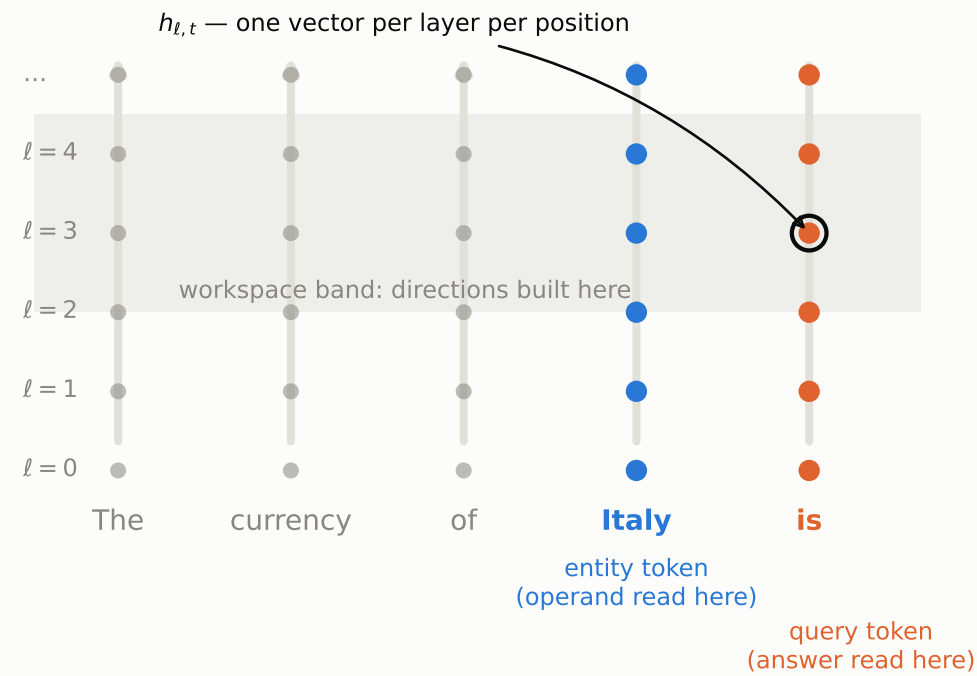


The state is a grid: layers × positions



One object, applied three ways

at each band layer ℓ , average the query-token states:

$$v_\ell(\text{op}) = \text{mean}_o [h_{\ell, -1}(o, \text{op}) - \text{mean}_k h_{\ell, -1}(o, k)]$$

≡ the ANOVA operator main effect $b_\ell(\text{op})$ (verified: dev. 10^{-8})

steer add $\alpha (v_\ell(B) - v_\ell(A))$ at the query token → margin flips; generates at calibrated α

compose replace $h_{\ell, -1}$ with $\mu_\ell + a_\ell(o) + b_\ell(k)$ → generates at the clean ceiling

decode classify / ANOVA the same states → factorization, layer profiles

Prompt → token × layer grid of residual states → per-layer operator direction by averaging at the query token → applied by addition (steering), by replacement (composition), or read out (ANOVA).